Embeddings to Words

Thursday, September 4, 2025 2:33 PM

Embeddings to Words

We have an LLM, words go in, turned into embeddings, then words come out

Numbers are easy because they're naturally ranked
Images are represented by pixels, which are comprised of numbers
    Similar data structure, no matter what the image is

Words don't have this consistent structure which makes them difficult to compare
We want to capture meaning numerically
We don't know how to do this ourselves
Machine learning models create this relationship for us

Each word is given a vector of weights
    King = 1.0, 1.0, 0.0   man = 1.0, 0.0, 0.0   woman = -1.0, 0.0, 0.0   queen = -1.0, 1.0, 0.0
    King - Man + Woman = Queen

Cosine Similarity to see which vectors are most similar

Embeddings are different from model to model

Specify random seeds when training so that a method is repeatable

Answer to most questions:
    "We'll talk about it next week"

The data structure encapsulates the semantics of words

Can just use computers for the group project today
    Fail gloriously